Stanley Nelavala

Generalist Software Engineer: Typed Backend, TypeScript, Postgres and AWS

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Huntington, IN | Open to relocate anywhere

Summary

Generalist software engineer who owns systems from design to release at a 10-person startup. At work I designed the signed service-to-service API every platform module uses, cut a deep catalog query from 285 ms to 0.20 ms with keyset paging, fixed a regular expression denial-of-service bug, built two web and mobile products in TypeScript, React and Python, and automated AWS cost and security controls. I like strong types (TypeScript, Rust, typed Python), PostgreSQL and code that stays correct under retries. M.S. in Computer Science (4.0 GPA). AWS Certified Solutions Architect, Associate.

Experience

Software Development Engineer Nov 2025 – Present
Mass Technologies, team of 10 (RxApps360 hospital platform, pre-launch; numbers from dev and test) *Huntington, IN*

- Designed, built and integrated into every platform module the signed service-to-service API (M2M): each call proves which service and tenant sent it. After a security audit, hardened it with a Redis rate limiter and replay protection.
- Replaced offset paging with keyset paging on the catalog device list: on a 328,000-device dev database, the query for row 150,000 fell from 285 ms to 0.20 ms. Built the search API on a shared service layer with a unique-id tie-break, tested with rows inserted, deleted and tied between pages.
- Cut the planning Items page from 3 API calls and 162 KB to 2 calls and 37 KB with typeahead search and cursor paging.
- Fixed a regular expression denial-of-service bug in a shared validation package: an 80-character hostile input took about 138 s; the new linear pattern matched the old results on 60,000 random inputs.
- Built RxDesign, the hospital design service, backend and web (FastAPI, React, TypeScript), with a deterministic, rule-based layout engine on a 0.5 ft grid; 220+ offline tests on the API.
- Built the RxAsset iOS and Android app (React Native, Expo) with scan-to-search and an on-device voice assistant; set up separate iOS and Android release pipelines in GitHub Actions with automatic TestFlight submission.
- Standardized malware scanning of file uploads (GuardDuty) across 9 app modules. Automated weekday start and stop of dev and acceptance servers and databases (ECS, RDS): about 53 hours a week instead of 168.

Adjunct Faculty (part-time) Jan 2026 – Aug 2026
The College of St. Scholastica *Duluth, MN*

- Taught Programming II (Java) and Machine Learning for Data Science.

Graduate Research Assistant and Teaching Assistant Aug 2023 – Aug 2025
University of Minnesota Duluth *Duluth, MN*

- Built a pipeline (Python, Node.js, MySQL, FFmpeg, PaddleOCR) that collected and analyzed 1M+ TikTok videos from 300k+ authors for thesis research on misinformation, then grouped themes with BERTopic; presented at the UMD CS graduate poster session.

Selected personal projects (public code, tests)

Card Dispute Service | *C#, .NET 10, PostgreSQL, EF Core, xUnit* GitHub

- Dispute state machine where two racing decisions on one case give exactly one winner (PostgreSQL xmin optimistic concurrency); idempotent create; tests written first.

Payments Ledger Service | *Java, Spring Boot, PostgreSQL, TypeScript, React* GitHub

- Double-entry ledger with idempotency keys and a transactional outbox; property tests and a load check that no money is created or lost.

Skills

Typed languages: TypeScript, Python (typed, Pydantic), Rust, Java, C#, Go, SQL
Web and APIs: React, Node.js, FastAPI, REST, GraphQL (local project), gRPC, signed service APIs, keyset paging, idempotency
Data: PostgreSQL (query tuning, migrations, optimistic concurrency), Redis, S3, SQS
AWS and delivery: ECS, RDS, GuardDuty, CDK (local project), Docker, GitHub Actions, Terraform, OpenTelemetry
AI-assisted work: Claude Code, Codex, opencode; I check agent output with tests and written acceptance checks

Education and Certification

M.S. Computer Science, University of Minnesota Duluth, 2023 – 2025, GPA 4.0, Graduate Summer Research Fellowship 2024 and 2025
B.Tech Computer Science and Engineering, IIIT D&M Kancheepuram, 2019 – 2023
AWS Certified Solutions Architect, Associate, Feb 2025 – Feb 2028 (verifiable on Credly)